

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 8

Aim: Performing Advanced Analytics using Trends, Clustering, and Forecasting

- a) Add and interpret trend lines (linear, exponential)
- b) Customize and analyze trend models
- c) Perform clustering on datasets
- d) Analyze distributions using built-in tools
- e) Apply forecasting techniques
- f) Compare different statistical models
- g) Export and interpret statistical results

Dataset: Indian_Climate_Dataset_2024_2025.csv

A: Add and Interpret Trend Lines (Linear, Exponential)

Task 1: Connect Dataset & Build a Scatter Plot / Line Chart

Open **Tableau**, load the CSV, and rename the sheet **Temperature vs Humidity Trends**. Then set **Humidity (%)** → **Columns**, **Temperature_Avg (°C)** → **Rows**, and disable **Aggregate Measures**.

The screenshot shows the Tableau Desktop interface with the following details:

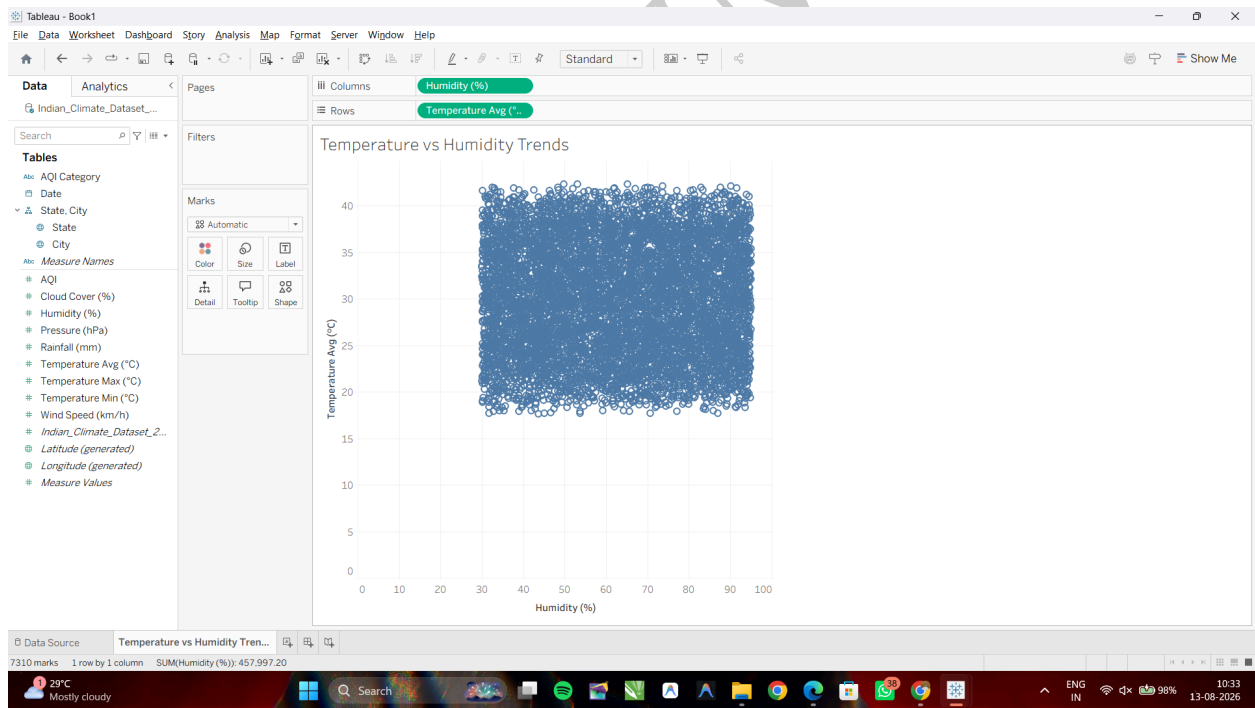
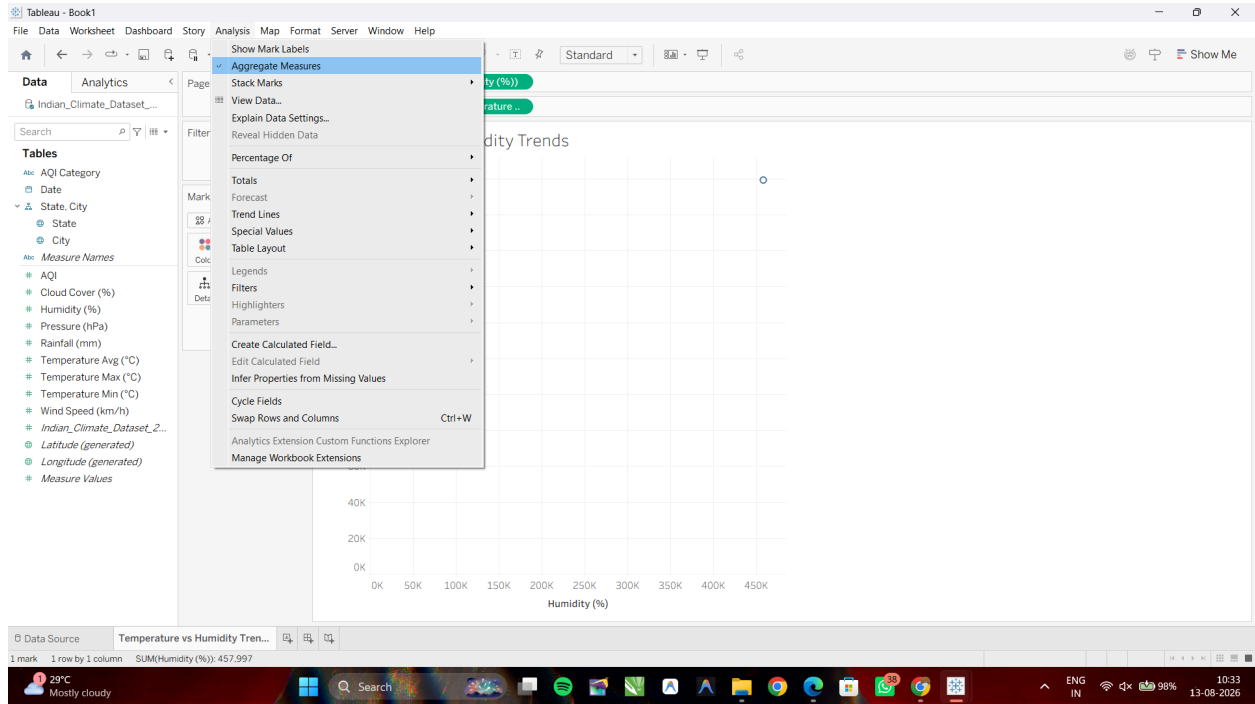
- Connections:** Indian_Climate_Dataset_2024_2025.csv
- Files:** A list of various CSV files is shown on the left pane.
- Data Source:** Indian_Climate_Dataset_2024_2025.csv
- Fields:** A table lists fields with their types and physical tables.
- Columns:** Date, City, State
- Rows:** Temperature Max (°C)
- Marks:** Automatic
- Data:** A table showing data for various cities and states, including Temperature Max (°C).

Type	Field Name	Physical Table	Rem...
Date	Date	Indian_Climate_Dataset_2024_2025.c	Date
City	City	Indian_Climate_Dataset_2024_2025.c	City
State	State	Indian_Climate_Dataset_2024_2025.c	State

Date	City	State	Temperature Max (°C)
01-01-2024	Mumbai	Maharashtra	32.5000
01-01-2024	Delhi	Delhi	25.4000
01-01-2024	Bengaluru	Karnataka	37.2000
01-01-2024	Chennai	Tamil Nadu	37.2000
01-01-2024	Kolkata	West Bengal	27.4000
01-01-2024	Hyderabad	Telangana	44.4000

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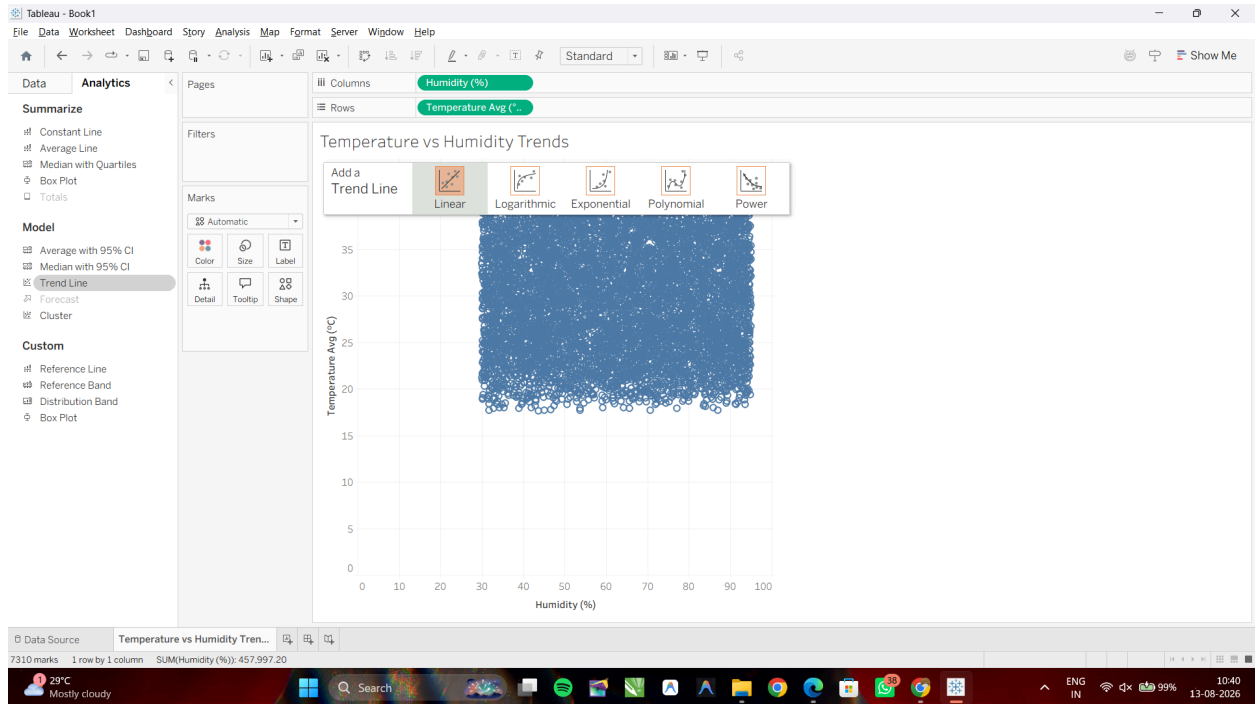
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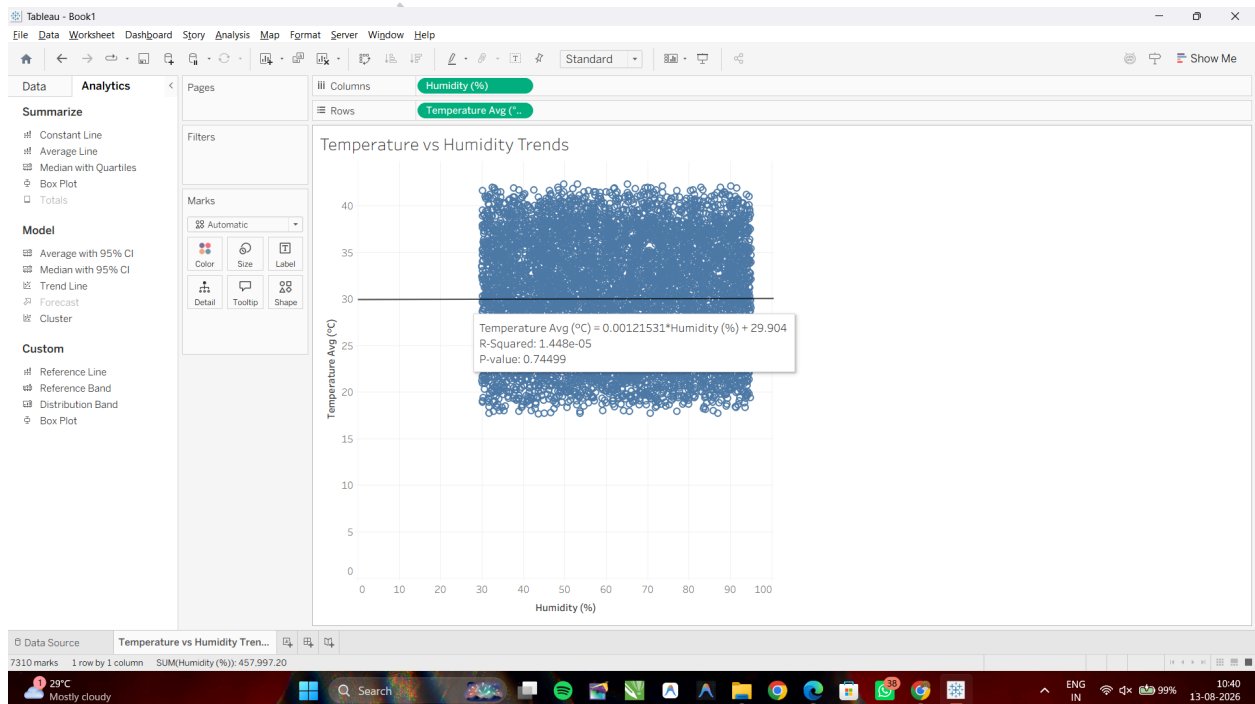
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Task 2: Add a Linear Trend Line

Open the **Analytics** pane, drag **Trend Line** onto the chart, and drop it on **Linear** to add a straight trend line.

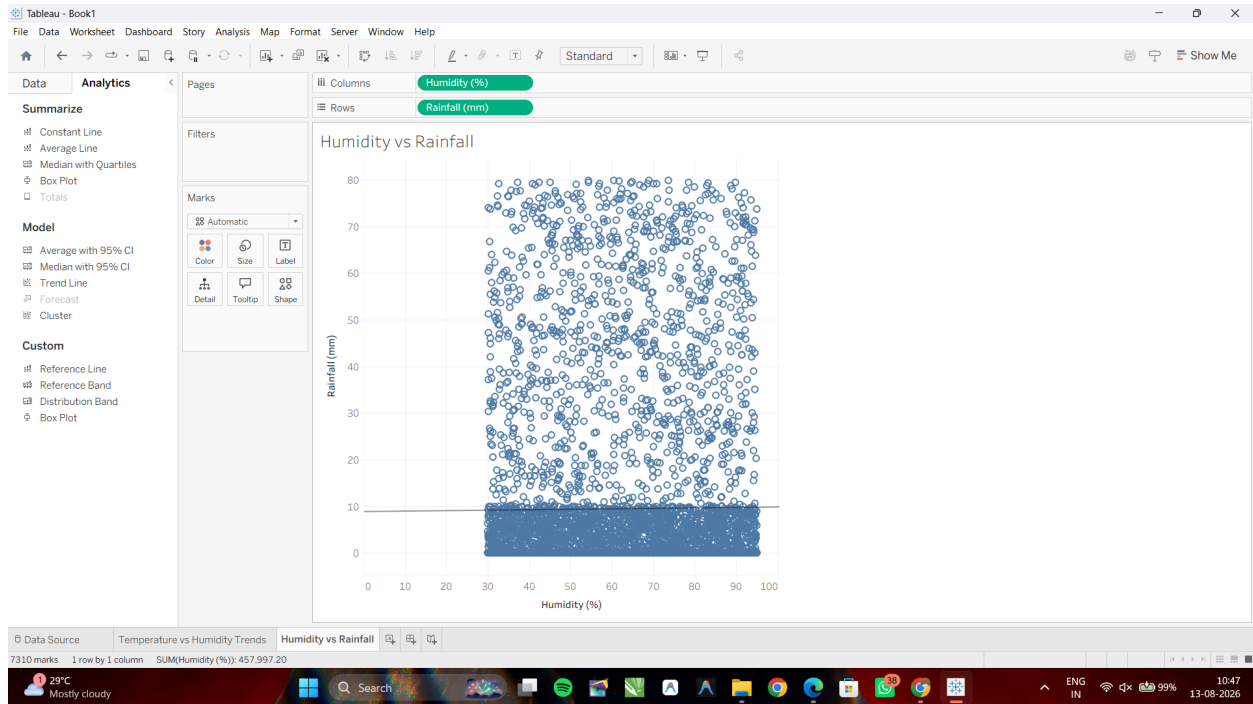


Hover over the trend line to view the **linear equation**, **R² value**, and **P-value** automatically provided by Tableau.



Task 3: Add an Exponential Trend Line for Non-Linear Measures (Rainfall vs. Humidity)

Create a worksheet named **Humidity vs Rainfall**, place **Humidity (%)** on Columns and **Rainfall (mm)** on Rows, uncheck **Analysis** → **Aggregate Measures**, then add an **Exponential Trend Line** from the **Analytics** pane.



Task 4: Interpret the Results

Linear Model: Shows a steady direct or inverse relationship, such as humidity increasing or decreasing average temperature.

Exponential Model: Shows a rapidly accelerating relationship, such as rainfall increasing sharply at higher humidity levels.

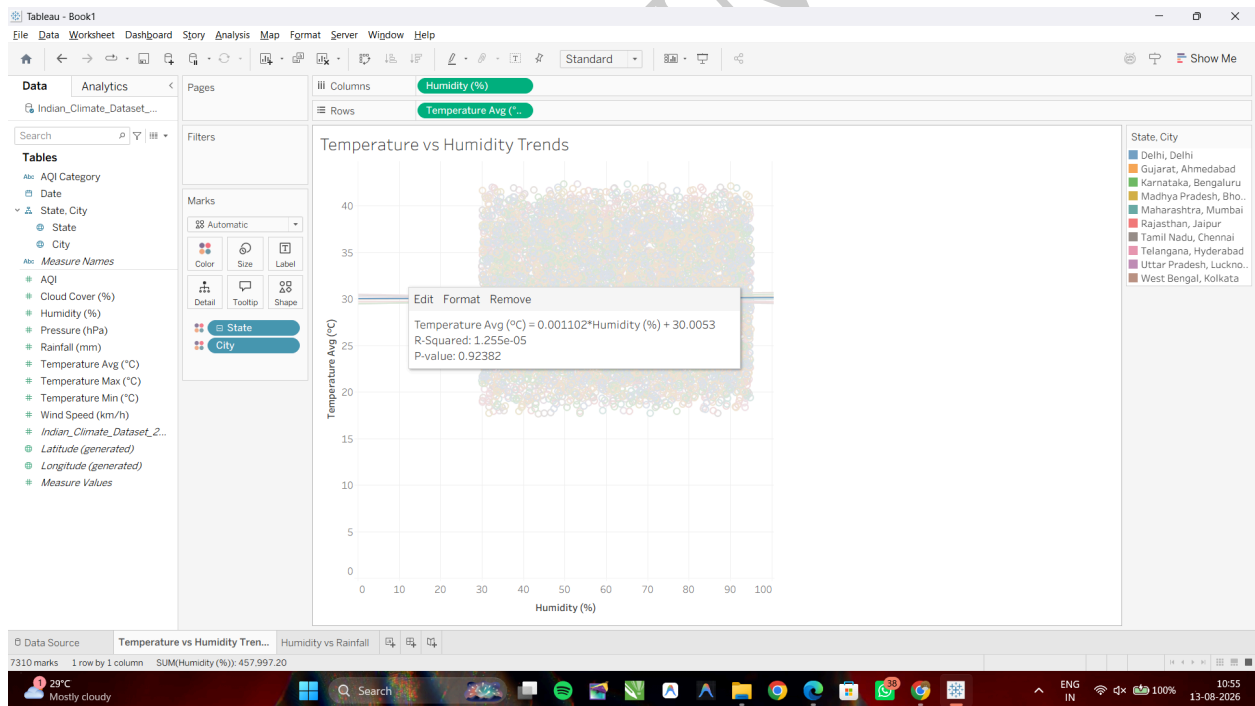
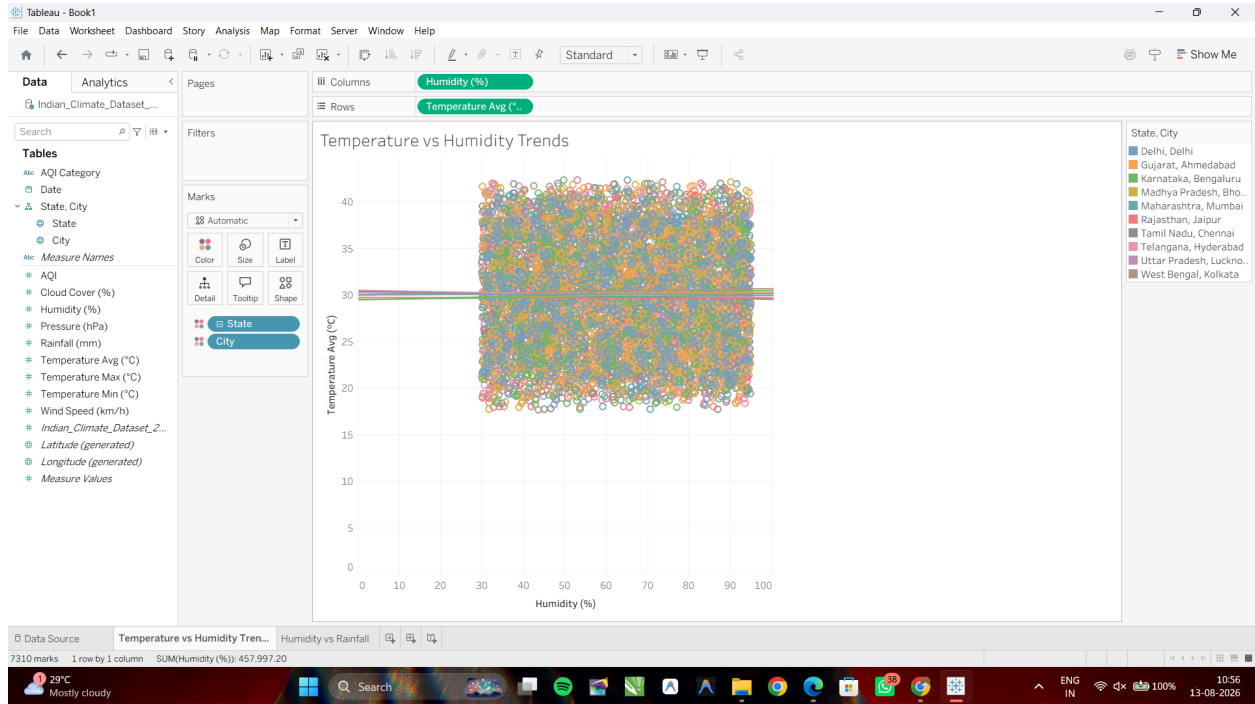
B: Customize and Analyze Trend Models

Task 1: Disaggregate Trend Lines by City

Open or duplicate **Temperature vs Humidity Trends**, drag **City** to **Color** on the Marks card, then compare the individual trend lines for coastal and inland cities.

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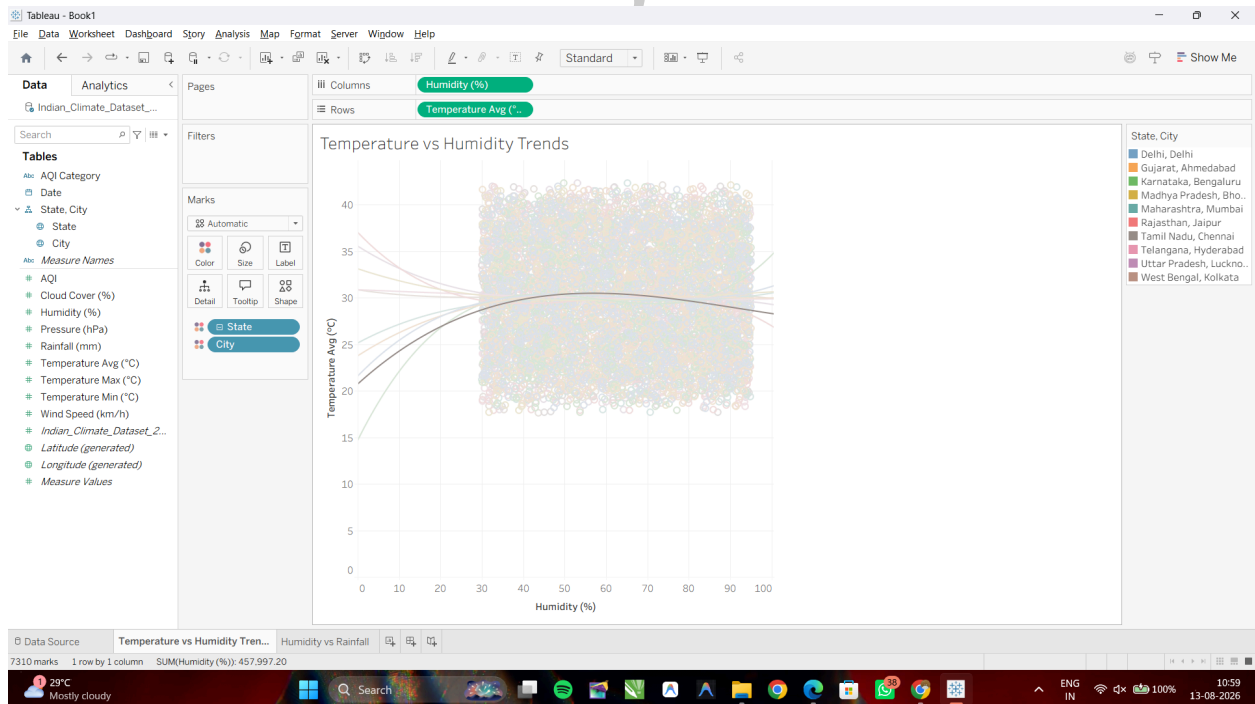
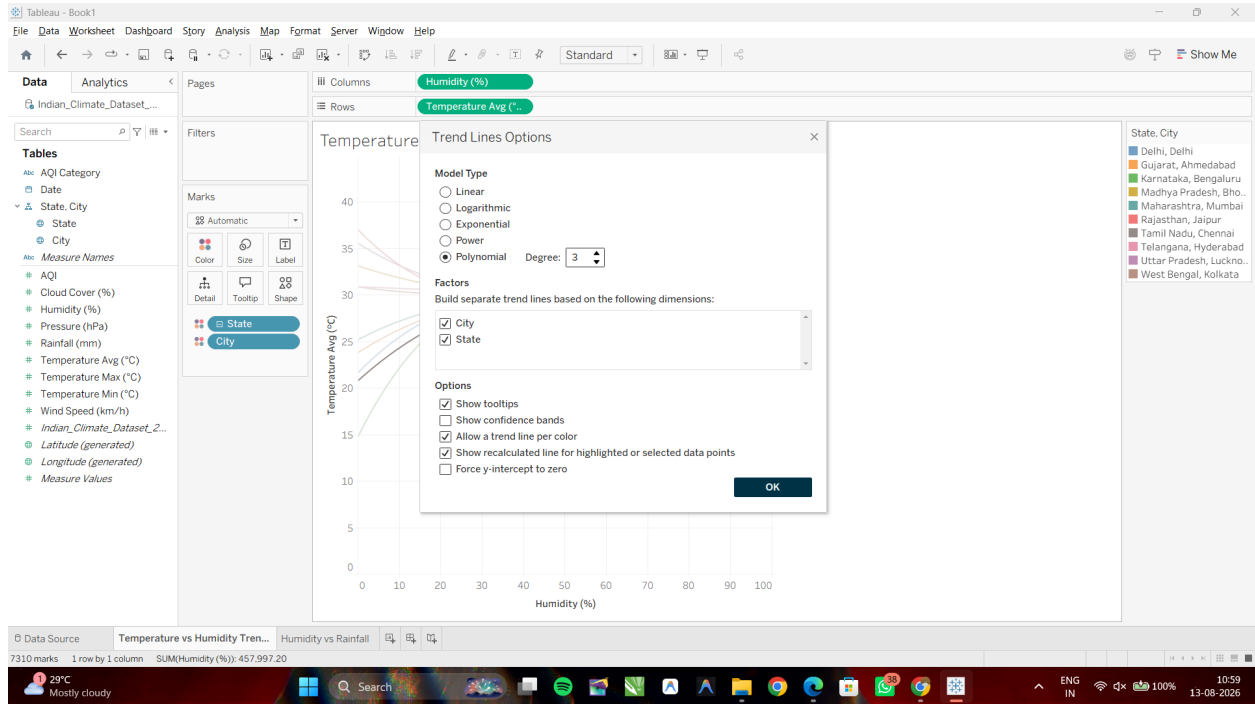


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Task 2: Compare Different Model Types (Polynomial & Logarithmic)

Right-click any trend line, select **Edit Trend Lines**, compare the **Linear**, **Logarithmic**, and **Polynomial** models, then choose **Polynomial (Degree 2)** and click **OK** to observe the curved fit.

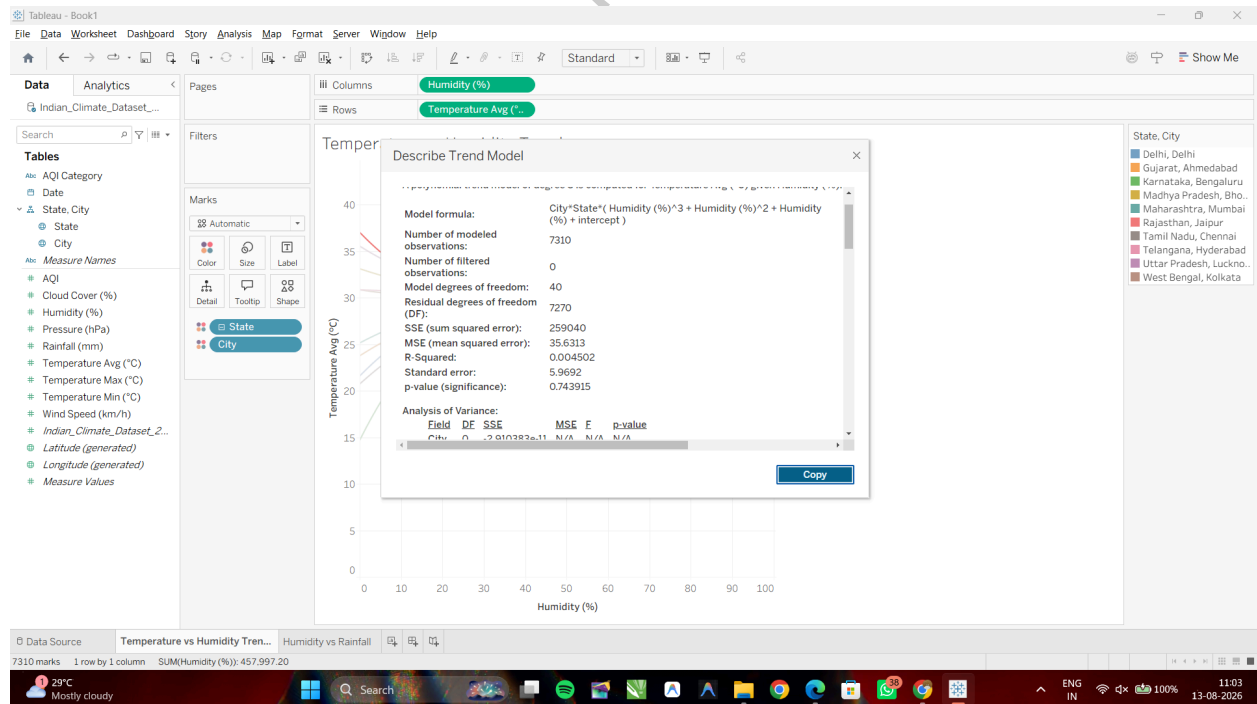
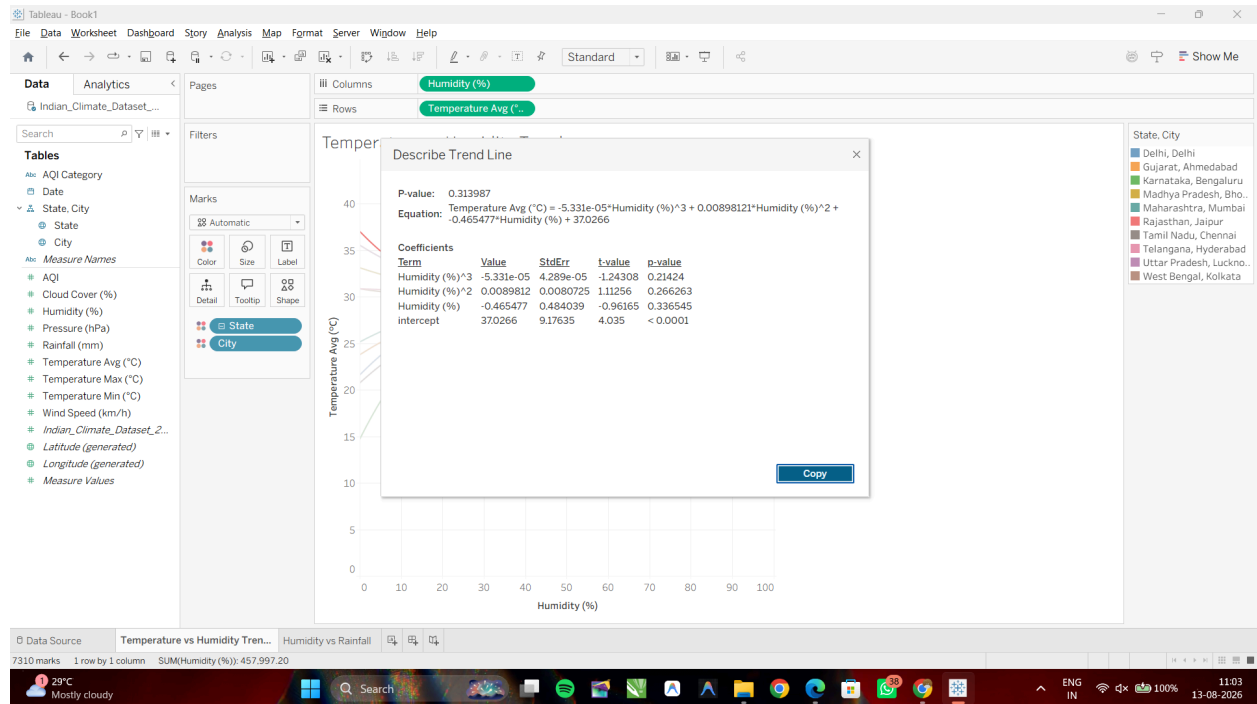


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Task 3: Access Full Statistical Summaries (Describe Model)

Right-click the chart and select **Describe Trend Line** to view equations, R^2 , and P-values, then select **Describe Trend Model** to review ANOVA, degrees of freedom, SSE, and F-statistic, and compare cities by their R^2 values.

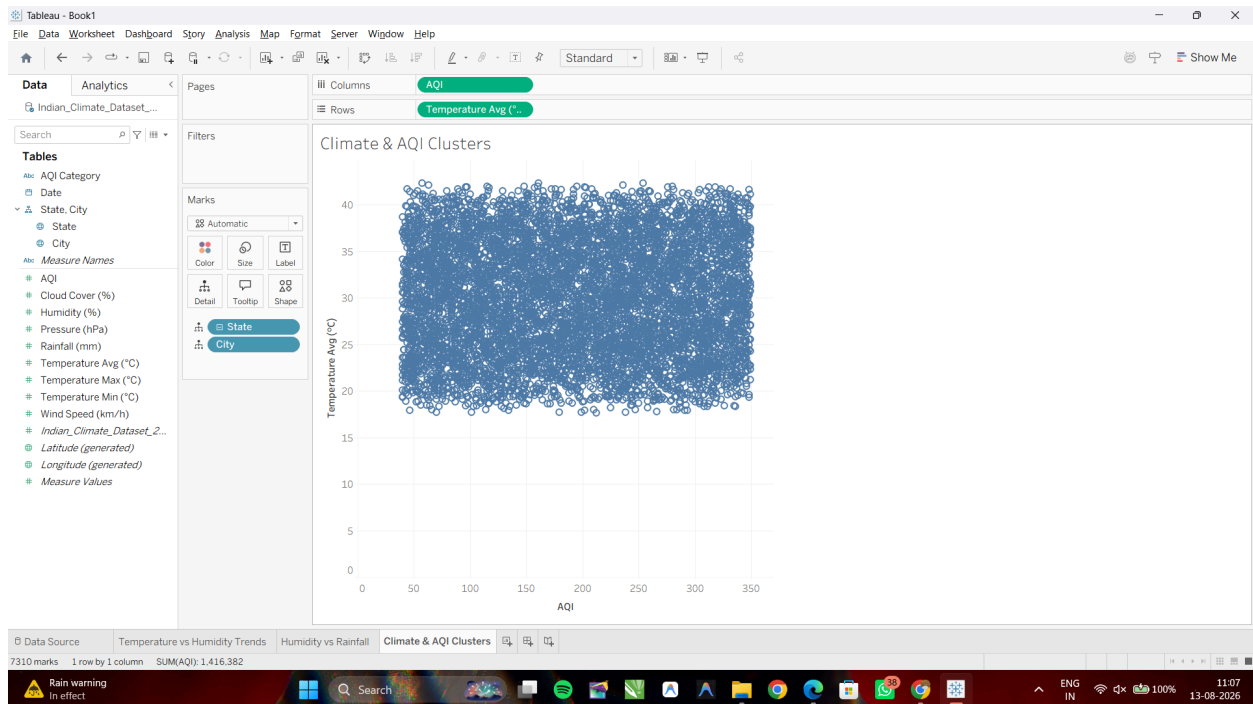


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C: Perform Clustering on Datasets

Task 1: Build a Multi-Metric Scatter Plot Matrix

Create a worksheet named **Climate & AQI Clusters**, place **AQI** on Columns and **Temperature_Avg (°C)** on Rows, uncheck **Analysis** → **Aggregate Measures**, and drag **City** to **Detail** on the Marks card.

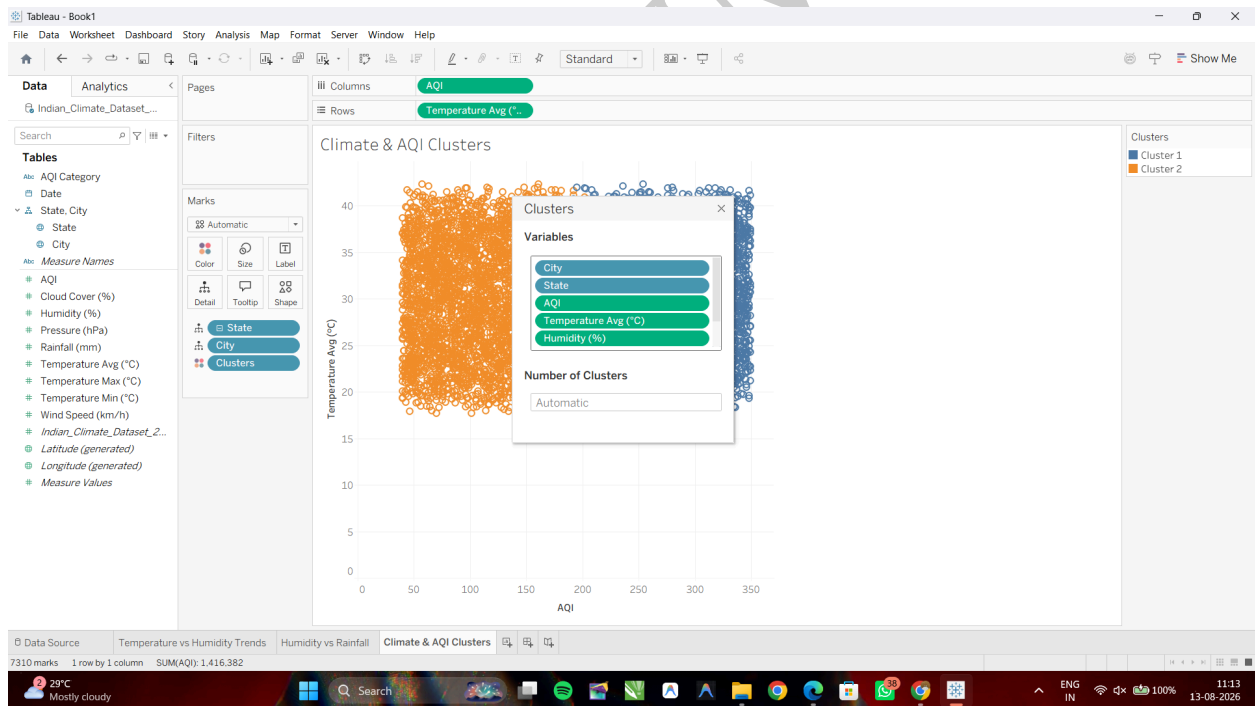
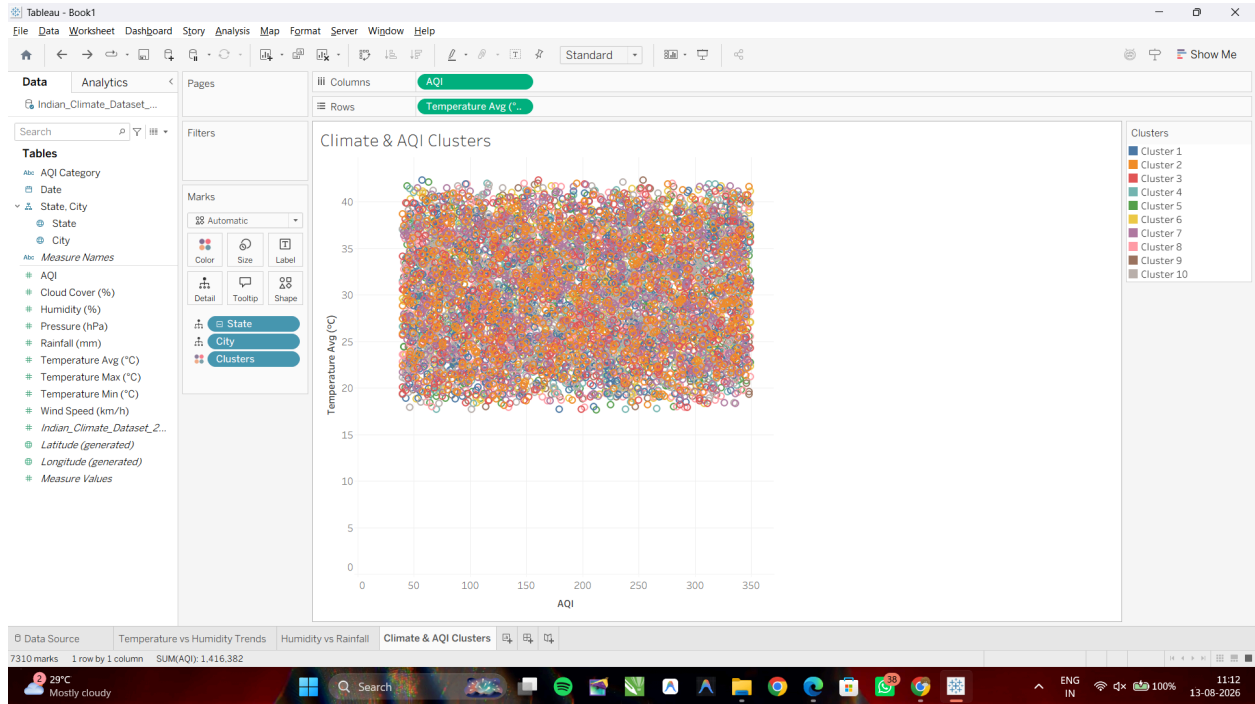


Task 2: Apply Built-in Tableau Clustering

Open the **Analytics** pane, drag **Cluster** onto the chart, add **AQI**, **Temperature_Avg (°C)**, **Humidity (%)**, and **Wind_Speed (km/h)** as variables, leave **Number of Clusters** as **Automatic**, and close the dialog to color-code the data points by cluster.

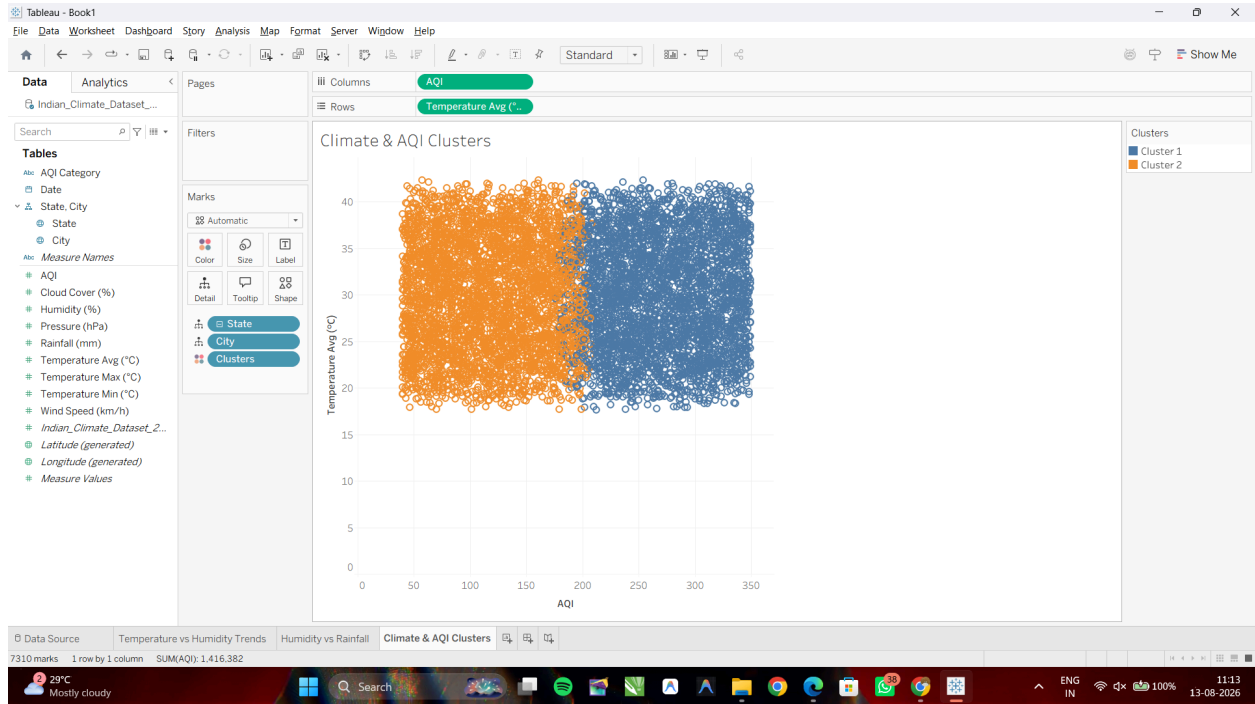
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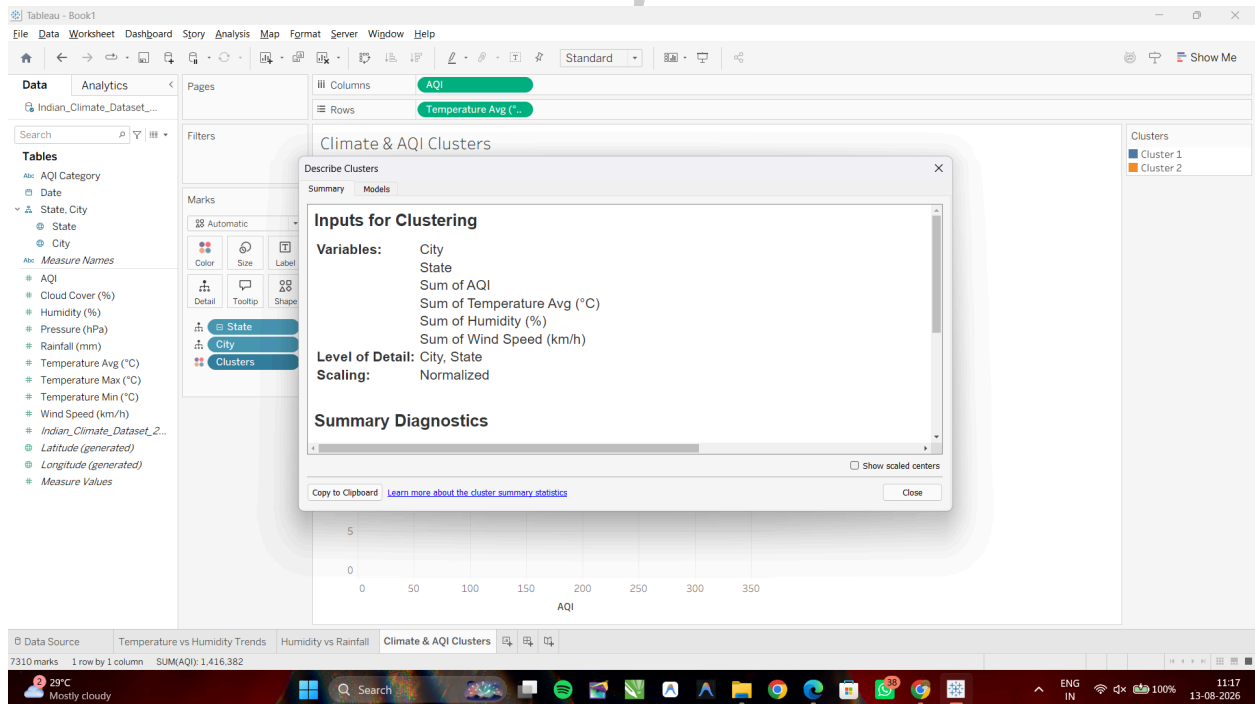
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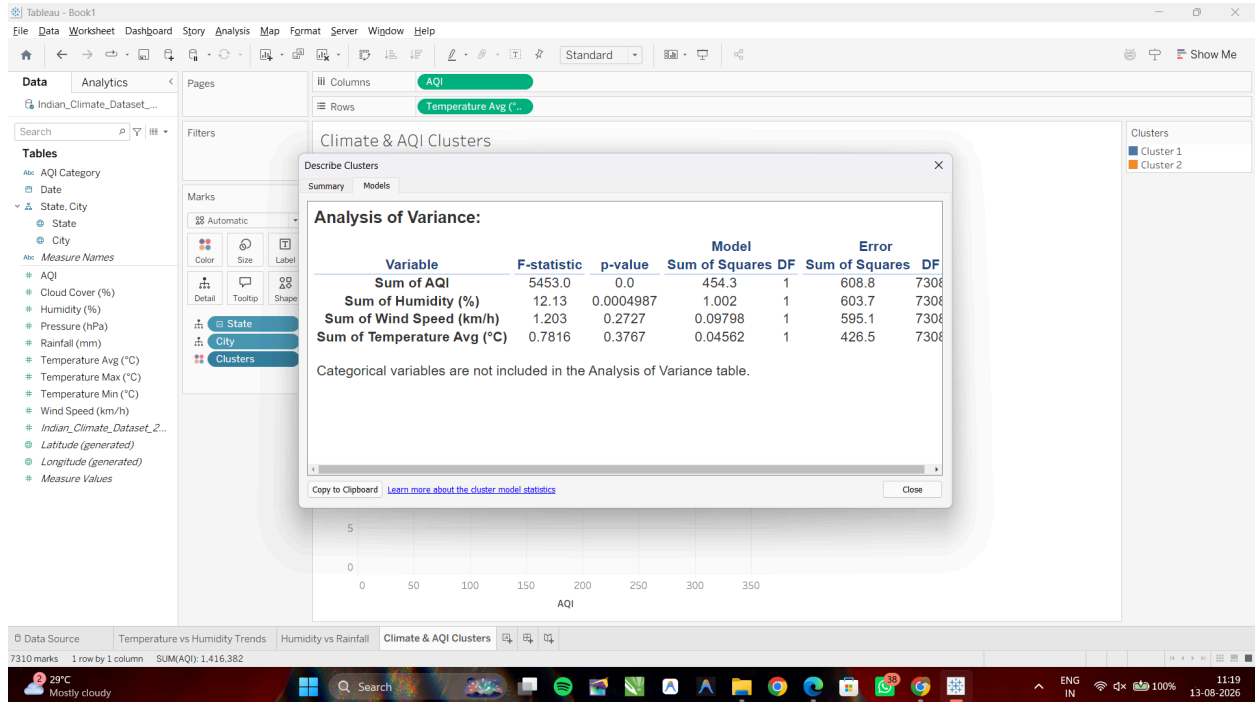
Task 3: Analyze Cluster Profiles & Save as Dimension

Right-click Clusters in the Marks card or legend, select Describe Clusters, then review Summary for each cluster's mean values and Models for ANOVA F-statistics and p-values showing which variables best separate the clusters.



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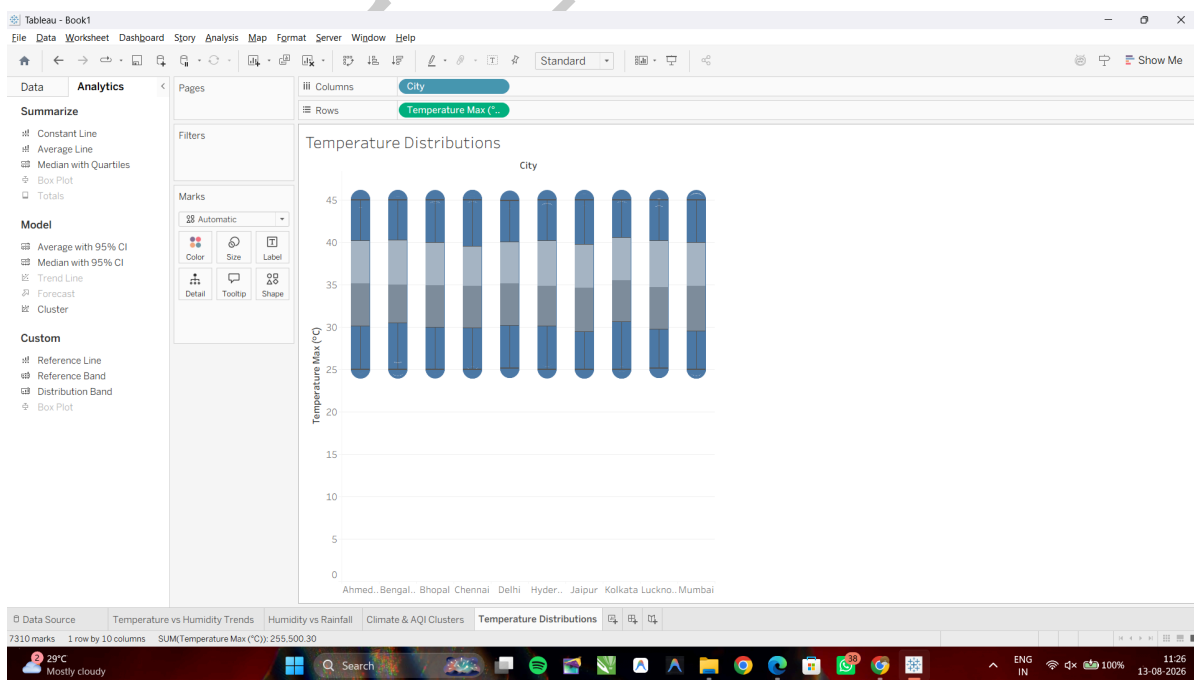
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D: Analyze Distributions using Built-in Tools

Task 1: Create a Box Plot for Temperature Distributions by City

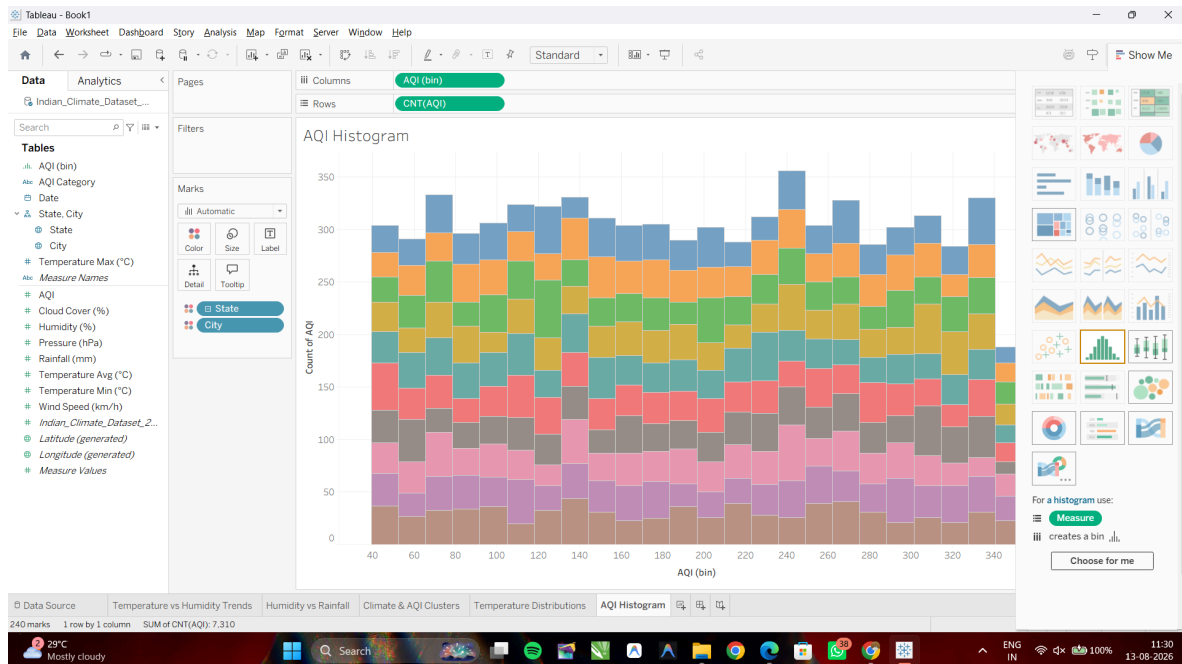
Create **Temperature Distributions**, place **City** on Columns and **Temperature_Max (°C)** on Rows, uncheck **Analysis** → **Aggregate Measures**, then use **Analytics** → **Custom** → **Box Plot** and hover over the plots to view **Q1**, **Median**, **Q3**, whiskers, and outliers.



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Task 2: Build an AQI Histogram & Quantile/Percentile View

Create **AQI Histogram**, select **AQI**, choose **Show Me** → **Histogram**, then drag **City** to **Color** on the Marks card to compare AQI distributions across cities.



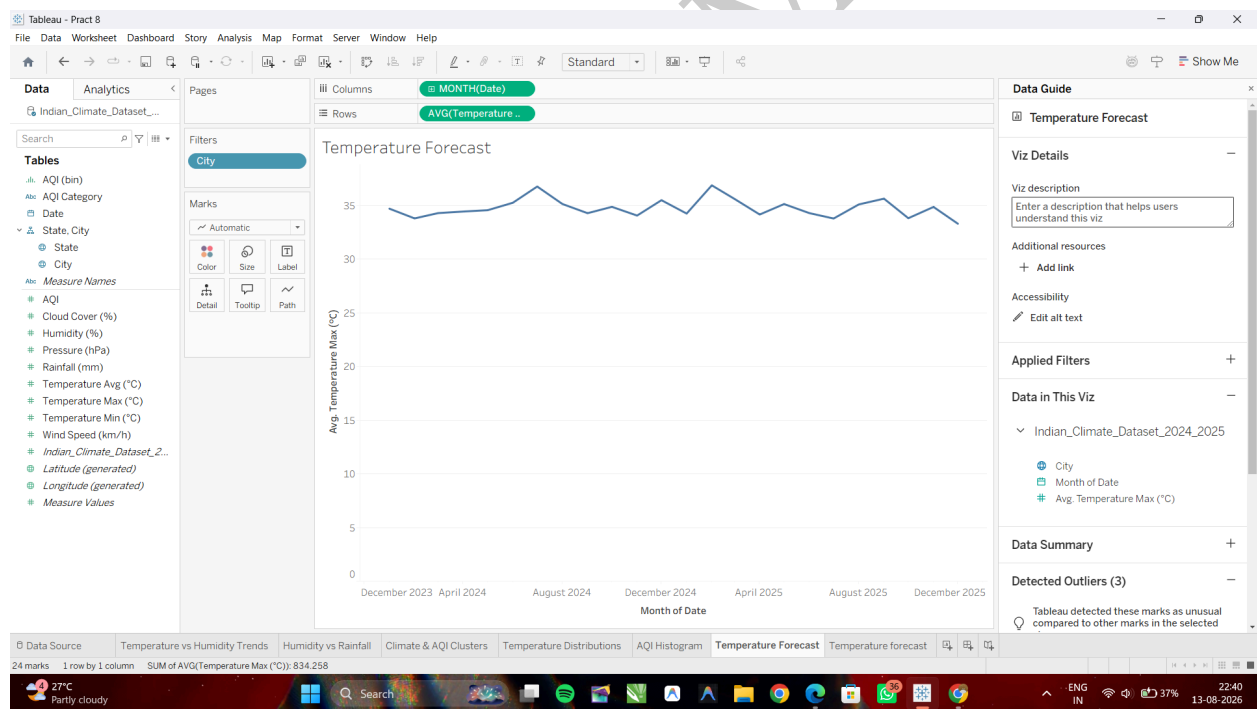
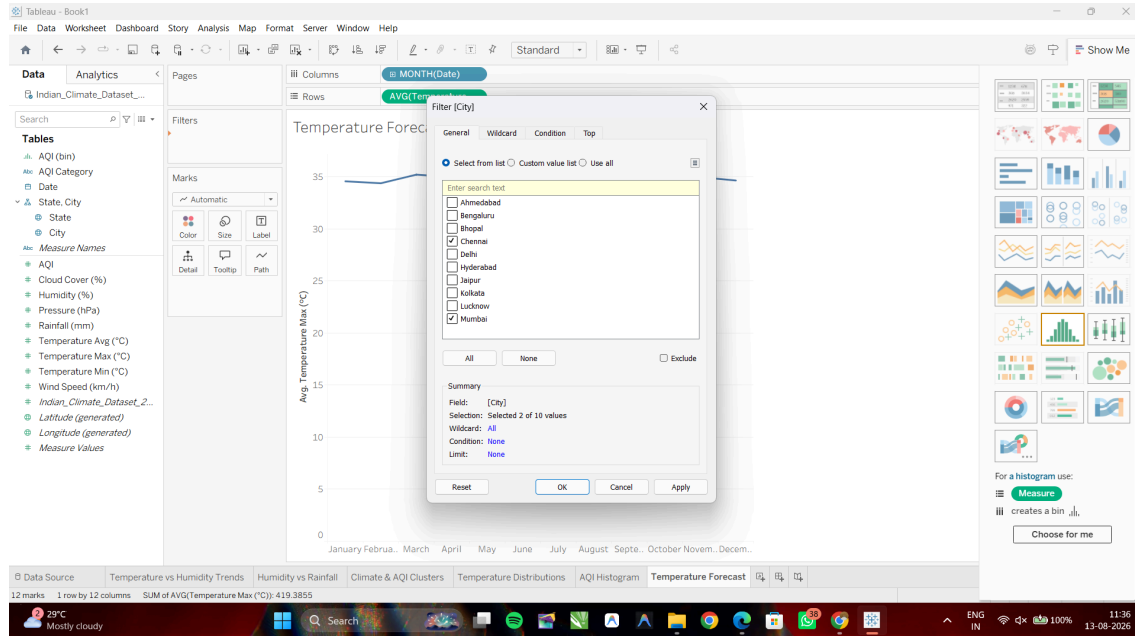
E: Apply Forecasting Techniques

Task 1: Set Up a Time Series Line Chart

Create **Temperature Forecast**, place **Date** on Columns as **continuous Month (Month Year)**, place **AVG(Temperature_Max (°C))** on Rows, and filter **City** to 2–3 cities such as Mumbai, Delhi, and Bengaluru.

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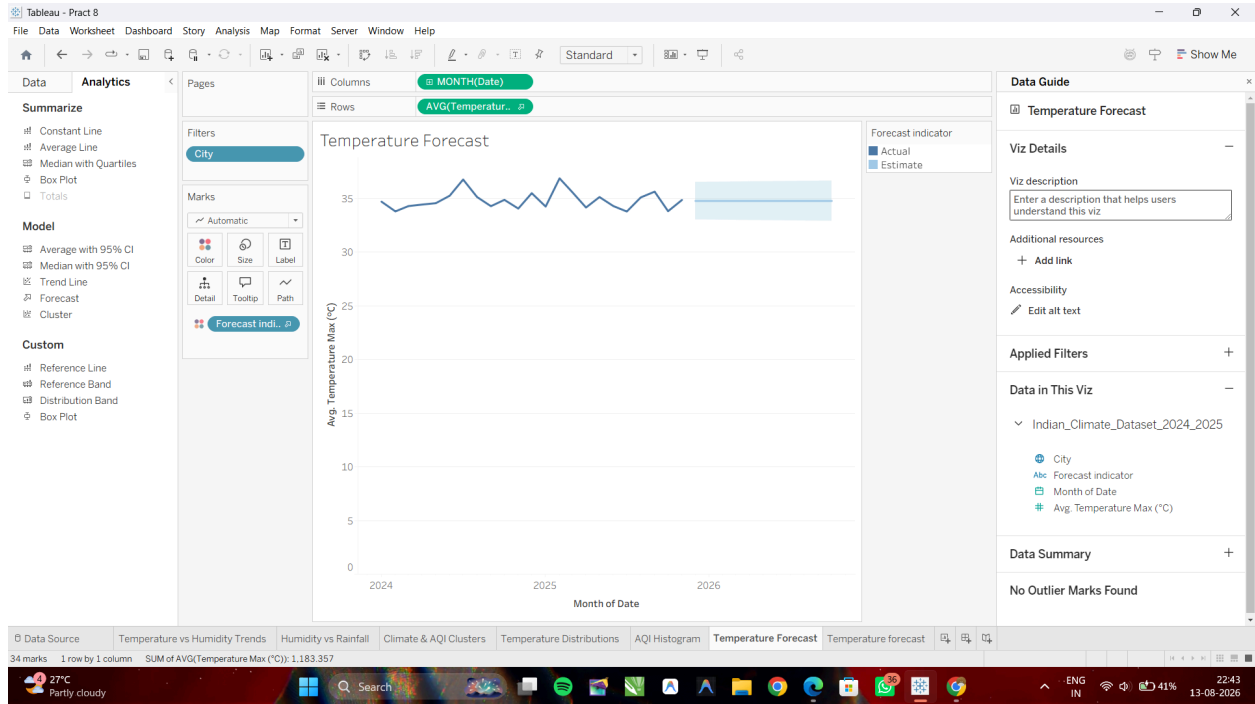


Open the Analytics pane, drag Forecast from Model onto the chart, and Tableau will extend the line with a shaded prediction interval showing the future forecast.

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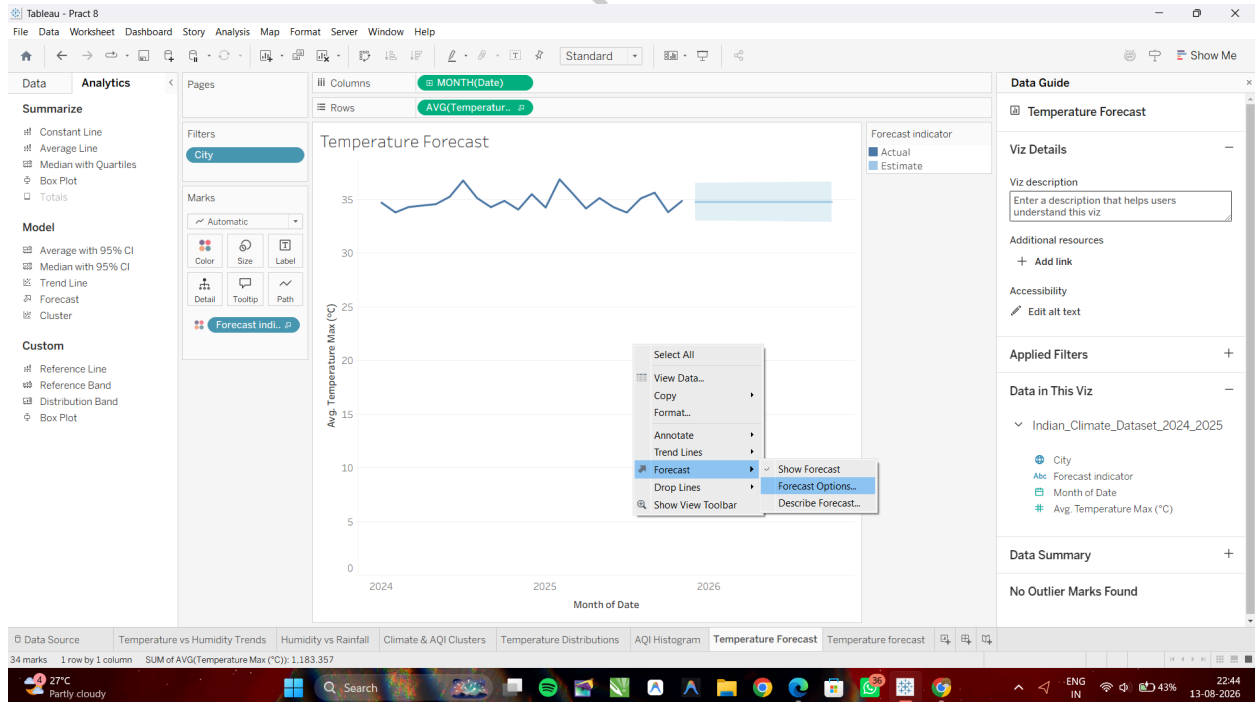
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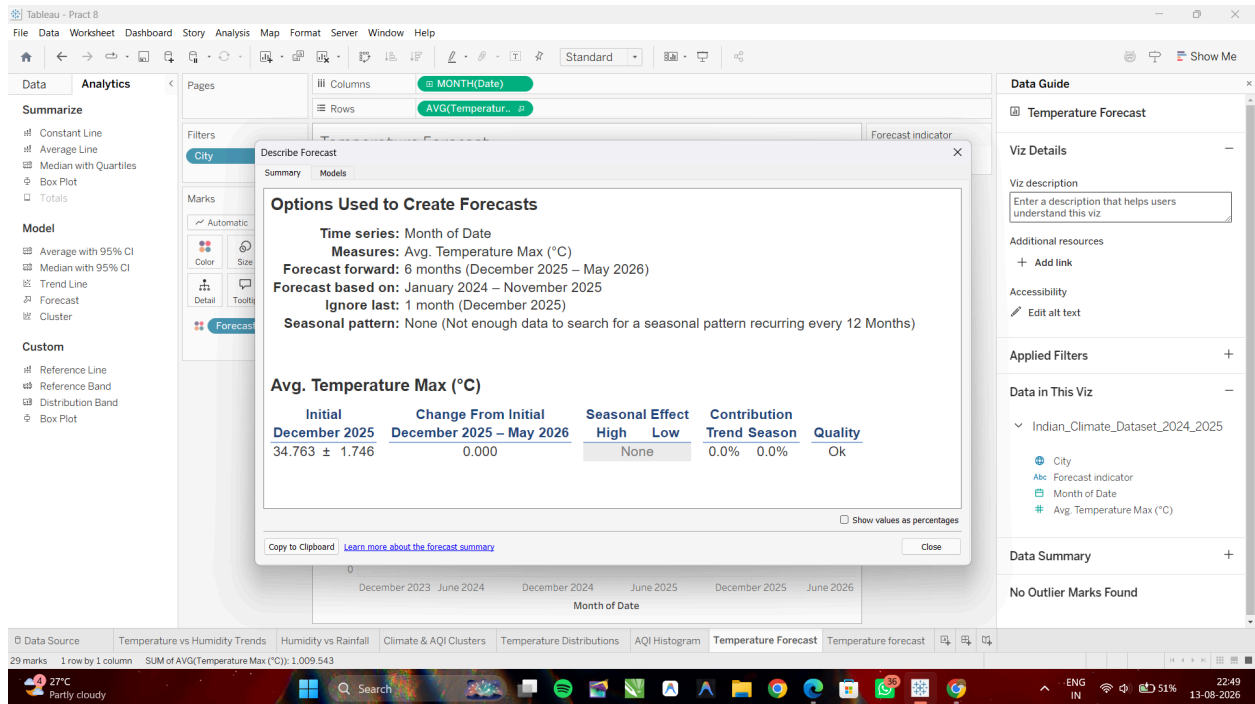


Task 2: Customize & Analyze the Forecast

Right-click the chart and go to **Forecast** → **Forecast Options**, set **Forecast Length** to **Exactly 6 Months** and **Prediction Interval** to **95%**, then use **Forecast** → **Describe Forecast** to review **RMSE** and the selected **Additive/Multiplicative** model.



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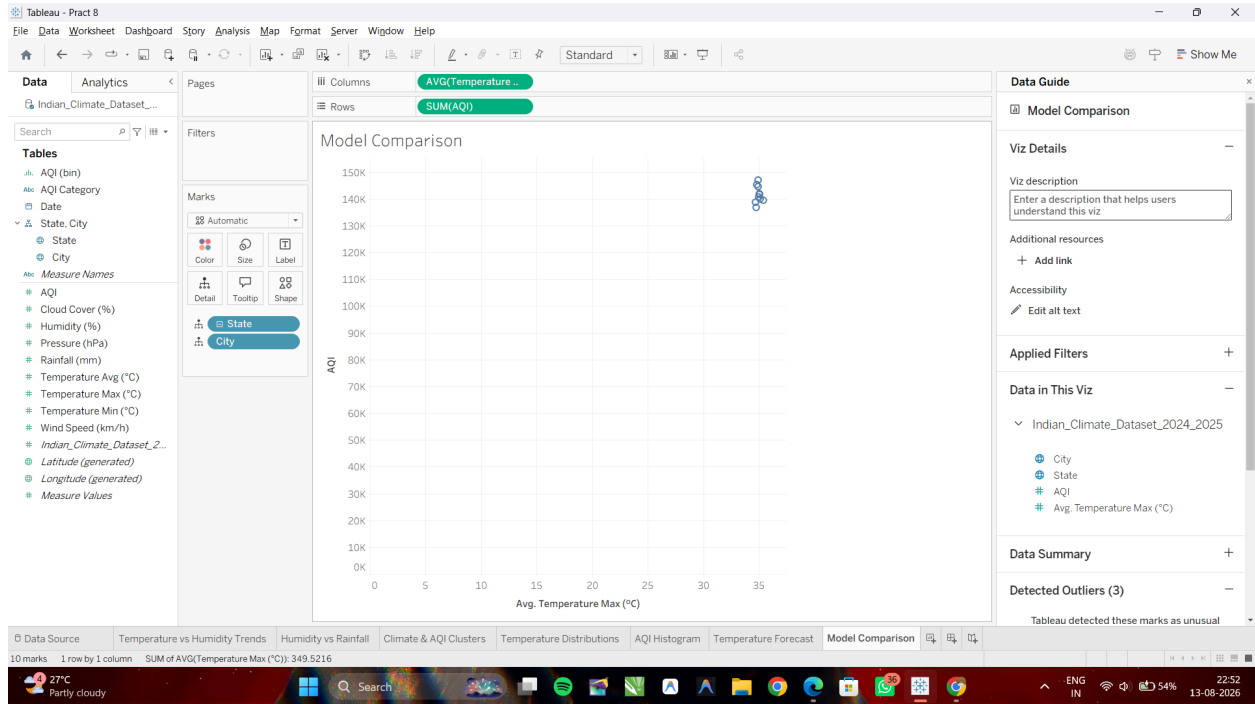
F: Compare different Statistical Models

Task 1: Set Up the Comparison View

Create **Model Comparison**, place **Temperature_Max (°C)** on Columns and **AQI** on Rows as continuous measures, then drag **City** to **Detail** on the Marks card to create the scatter plot.

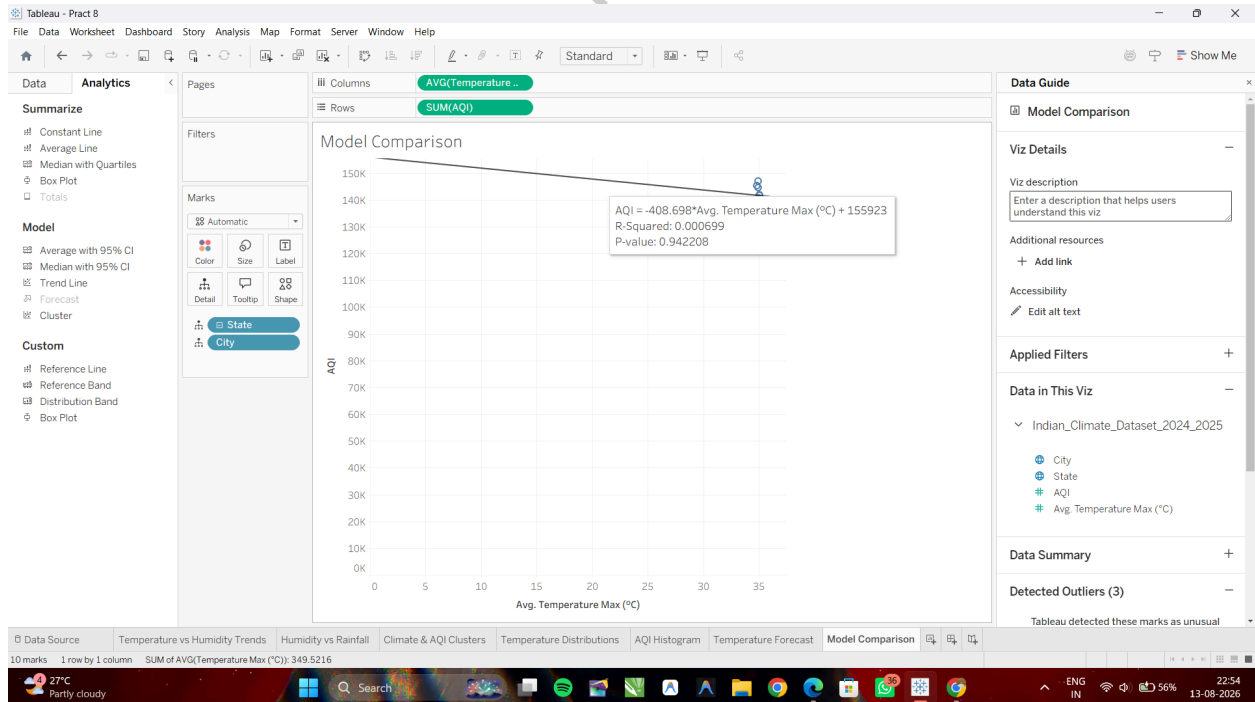
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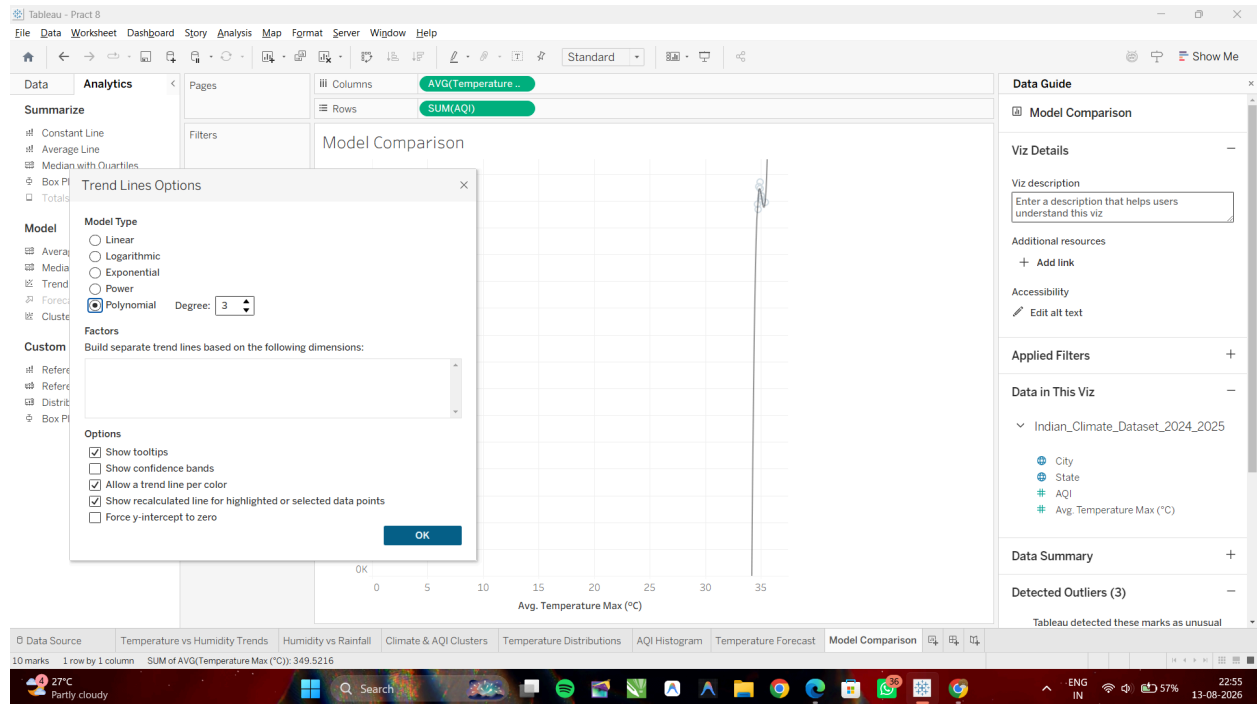


Task 2: Add and Swap Statistical Models

Open **Analytics**, add a **Linear Trend Line**, record its **R²** and **p-value**, then use **Edit Trend Lines** to test **Linear**, **Logarithmic**, **Exponential**, and **Polynomial** models and compare their **R²** and **p-values** to identify the best fit.



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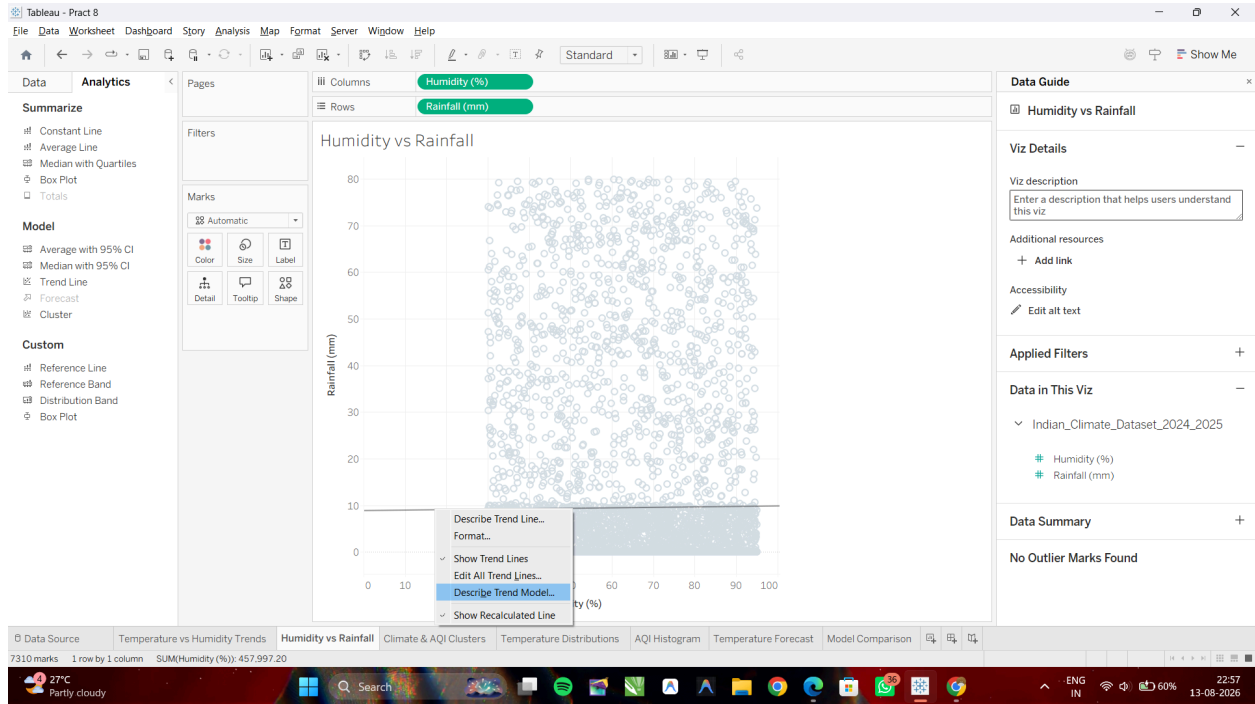
G: Export and Interpret Statistical Results

Task 1: Export the Statistical Model Description

Right-click the trend line or canvas, select **Describe Trend Model** (or **Describe Forecast**), then click **Copy** or **Export** to save the **ANOVA summary and parameter estimates** for your practical record.

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Task 2: Interpret the Key Output Metrics

When reviewing the results, focus on the **Equation Parameters** (intercept and slope), **R²** (the percentage of AQI variance explained by temperature), and the **p-value** (a value below **0.0001** indicates a statistically significant relationship).

